

Venkatesh Tantravahi

Utica-NY | (586)-742-1300 | vtantravahi@gmail.com | [linkedin.com/in/venkateshtantravahi](https://www.linkedin.com/in/venkateshtantravahi) | venkateshtantravahi.netlify.app

Professional Summary:

Data professional with 3+ years of experience in data engineering, analytics, and machine learning solutions. Proficient in Python, SQL, and cloud-based data workflows (AWS). Skilled in building scalable data pipelines, developing machine learning models, and creating impactful dashboards using Tableau and Power BI. Passionate about transforming data into actionable insights and automating decision-making. Currently pursuing an M.S. in Computer Science with a focus on AI and data science. Eligible to work in the U.S. under F-1 OPT STEM Extension.

Skills:

- **Programming Languages:** Python, SQL, R.
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI.
- **Machine Learning & Statistical Modeling:** scikit-learn, TensorFlow, Keras, Regression, Classification, Clustering, A/B Testing.
- **Cloud & Data Engineering:** AWS (S3, Lambda, Glue, Redshift, SageMaker), Docker, Git.
- **Databases:** MySQL, PostgreSQL, MongoDB.
- **Other Tools:** Jupyter, GitHub, Linux.
- **Soft Skills:** Analytical Thinking, Problem Solving, Communication, Collaboration.

Professional Experience:

Capgemini, Hyderabad (India)

Jun 2021 – Dec 2022

Role: Data Engineer & Analyst

Responsibilities:

- Designed and optimized SQL-based ETL pipelines and CDC frameworks using AWS Glue and Redshift, enabling real-time reporting for 500+ users.
- Built and deployed machine learning models (scikit-learn, TensorFlow) on AWS SageMaker to support predictive analytics and reduce data latency by 25%.
- Created interactive dashboards in Power BI, integrating AWS RDS and Redshift data sources for real-time business insights.
- Led data governance implementation (RBAC, Row-Level Security) in Power BI, ensuring secure access for 500+ stakeholders.
- Migrated on-prem SQL workloads to AWS RDS/Redshift, reducing query execution times by 10%.
- Developed CI/CD pipelines using GitHub Actions, Jenkins, and AWS Lambda to automate model deployments and improve production stability by 40%.
- Conducted A/B testing to measure dashboard impact, enabling 18% faster decision-making across teams.

Jukshio, Hyderabad (India)

Apr 2020 – Jun 2021

Role: Data Scientist

Responsibilities:

- Developed and deployed facial recognition models using TensorFlow and Keras on AWS SageMaker, achieving 98% accuracy and improving inference speed by 20%.
- Built end-to-end MLOps pipelines (AWS S3, Glue, SageMaker, Lambda) to automate model training, deployment, and monitoring, reducing downtime by 35%.
- Designed an identity verification system using pre-trained transformer models (BERT, RoBERTa) for document verification and anomaly detection, enhancing classification accuracy by 18%.

- Applied model interpretability techniques (SHAP, LIME) to improve transparency in AI-driven decision-making and build stakeholder trust.
- Developed a scalable REST API for model inference using Flask and Docker, deployed on AWS ECS and Fargate for secure, high-availability service.
- Automated infrastructure deployment using AWS CloudFormation for reproducibility and scalability.
- Conducted A/B testing for model versions, increasing stakeholder confidence and accelerating deployment cycles.

3IceCubes, Remote

May 2019 – Dec 2019

Role: Backend Data Engineer

Responsibilities:

- Designed and developed REST APIs for data ingestion and processing using Django and PostgreSQL on Google Cloud Run, improving data pipeline efficiency by 12%.
- Implemented secure authentication and authorization using Firebase Authentication with JWT, reducing unauthorized access incidents by 95%.
- Built data retrieval and transformation services for real-time client applications, enabling faster access to processed data.
- Deployed microservices architecture using Docker and Kubernetes on Google Kubernetes Engine (GKE), ensuring scalability and high availability.
- Automated CI/CD pipelines with GitHub Actions and Cloud Build, reducing deployment cycles by 40%.
- Created unit and integration tests using PyTest and Google Cloud Testing, catching 85% of critical bugs pre-deployment.

Projects: For more projects: [GitHub](<https://github.com/venkateshtantravahi>)

Forward-Forward Algorithm for Neural Networks

Explored an alternative learning method to backpropagation for deep learning models

- Implemented and extended the Forward-Forward (FF) algorithm to train neural networks without backpropagation, applying it to complex datasets with diverse activation functions.
- Conducted experimental evaluations comparing FF with standard backpropagation, analyzing training efficiency, convergence behavior, and generalization performance.
- Documented findings and contributed to AI research on alternative learning strategies, showcasing deep learning proficiency and experimental rigor.

SQLite Playground

Built an interactive SQL learning platform with real-time query execution and visualization

- Developed a web-based SQL editor using Streamlit and Python, enabling users to write, execute, and visualize SQL queries on sample datasets.
- Integrated syntax highlighting, auto-completion, and real-time error detection for enhanced user experience.
- Added data visualization features using Matplotlib and Plotly to generate graphical insights from query results, promoting data exploration.

DeBERTa Hate Speech Detection

Developed an NLP model to classify hate speech and offensive language in text data

- Fine-tuned a DeBERTa transformer model on labeled text datasets, achieving high accuracy in detecting hate speech and offensive content.
- Implemented model interpretability techniques using LIME and SHAP to ensure transparency and explainability of AI decisions.
- Designed and deployed a user-friendly API for inference, enabling seamless integration into applications requiring content moderation.

Education:

- **Master of Science**, Computer and Information Science | State University of New York, Utica, NY | Dec 2024
- **Bachelor of Technology**, Electrical and Electronics Engineering | Mahatma Gandhi Institute of Technology, India | 2020